

Vehicle Framework

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Frequently Asked Questions

Does/will the Vehicle Framework have _____ feature?

I would prefer not to answer questions like these since they pop up so frequently. These are the major features I have planned:

- Raiders / Traders (NPC's utilizing vehicles)
- Aerial Combat (Anti air defenses, S2A missiles, strafing + bombing runs, paratroopers / drop shock mechanics, combined tactics such as bombing before land forces arrive, etc.)
- Modular Upgrades
- Immersive World Events (related to roadside ambushes, infantry tactics, naval combat, etc.)

Why are there no vehicles?

This is a framework, not a content mod. There aren't any vehicles in the framework, nor will there be in the future. This is intentional, please do not ask for me to add any type of vehicle, the answer is no.

Are vehicles just reskinned pawns?

Yes and no. I am using a Pawn subclass so that caravaning and some other specific mechanics are easier to work around, but almost all pawn components have either been stripped or replaced. They are not the same "pawn" vehicles you see in other mods. They are actually closer to buildings.

Can colonists use vehicles to get around faster on the map?

Not from AI behavior, no. While the vehicles are usable, colonists won't use them to goto locations quicker (without the player telling them as much). They're mostly for caravan usage or combat. Will this be implemented in the future? It's likely but we'll see, there are a few technical limitations to consider.

Will other Factions make use of Vehicles?

Yes, vehicles will show up in all kinds of areas including raids, trader caravans, and base defenses. Vehicle generation for NPCs has already been completed and AI behavior for turrets is also done. So not right now ... *but it is planned.*

Vehicles will not have their own unique raids or traders. They will be injected into existing incidents to replace overwhelming numbers (helps improve performance) and give a bit more variety to the game. The part that needs to be done is how the vehicles will path, what priorities they will have, and how the raiders' AI should be changed for ground troops. I want modders to have control over where they show up, but I don't want them to be some rare unique raid (like

the full sniper raid). They will be fairly common when a map's wealth value reaches a certain threshold.

Trains?

I won't be implementing trains. I'm sure it wouldn't be that hard for some other modder to do. I've made the (open) source code fairly accessible to other modders.

Will shuttles be able to fight on the world map (aka dogfights)?

Not yet. It would be cool but it's just something way out of scope for what vehicles should be adding. Maybe later.

Will shuttles be able to bomb or strafe like in SRTS?

At some point in the future, yes. After I've finished adding raider / trader AI for using vehicles, I will begin working on bombing and strafing runs as well as giving raiders the capabilities of using them against your base before a raid. Some ideas I've also had to go alongside bombing and strafing are:

Air lifts (drop cargo over a map without having to land)

Paratroopers (drop pawns over a map without having to land, animation may be parachute style, or even drop pods ODST style)

Missiles for local and world maps.

Can Pawns shoot their weapons from vehicles?

No. It's not as easy to implement properly as you'd think considering the AI would also have to take this into account, rendering on top of the vehicle would look janky, and it would be really difficult to balance.

Unmanned Vehicles?

Yes, although "unmanned" and "AI controlled" are not the same. Unmanned vehicles will be player controlled but won't require colonists to fill up certain roles in order to drive. You'll still need to give them orders though.

Are there limited traverse angle weapons?

Yes, for vehicle turrets there is a field for specifying restricted angles in which the weapon can rotate. On top of this you can also set your turret to "static" which will not allow targeting and just fire off the turret in whatever direction it is pointed in (you can set the default angle of which it is pointing relative to North).

Furthermore, you can even attach turrets to other "parent" turrets which will still enforce the restricted angles. This is how the mounted machine guns work for the example Tank I use for

development. The turrets rotate with the main turret but are still restricted by a 45 degree cone in whatever direction the main turret (and vehicle) is facing.

Can I run over enemies with my car?

Yes, there is such a feature however I'm not a huge fan of it. Some vehicles are just too fast for pawns to be able to get out of the way in time after being notified of danger. Even with plenty of notification time, canceling / reassigning prioritized jobs for a goto job must wait until the pawn has fully transitioned to the next cell which may not leave enough time for the pawn to move out of the way. Friendly fire is on? Great, you just killed 2 colonists and your chihuahua while trying to leave the map for a caravan. Friendly fire is off? You no longer have any challenge while you mow down raiders in droves with a Volkswagen. It's a tough feature to balance.

How does _____ feature work?

Read the google doc. If it's not yet on the google doc, bug me about it on my discord server and I'll be sure to write a section for it. The google doc lags behind feature implementations so feel free to give me a ping if something is missing.

Can vehicles be modular?

That's part of the upgrade system I created, but it is disabled at this time until I can fix some of the bugs in it. I'm the only one working on this project so in order to not spend an eternity on the mod, I've had to put off certain features for later. It's pretty much done but I need to spend a bit of time cleaning up the UI, testing, and then fixing bugs.

Will there be upgrades for visual modifications?

Yes, alongside other upgrades, the base upgrade node class will have an option to add a texture that will be layered with the vehicle when rendered.

Do pawns fulfill needs like rest and hunger while inside vehicles?

Yes for both in local maps and on the world map.

Will there be support for multiple turrets?

Yes, you can add an unlimited number of turrets to your vehicle (as a modder). As a player you're restricted to whatever turrets are available for that vehicle.

What will happen to my pawns inside the vehicle when it's destroyed?

There are "roles" for each vehicle that represent where the pawn is and what their job is. That pawn will likely take damage when damage to the vehicle has penetrated, meaning internal parts are now being hit. When this happens, there is a significant chance for one of your pawns to be hit instead. When the vehicle's HP hits 0, all surviving pawns, corpses, and inventory items will be ejected. If the vehicle is over impassable terrain (eg. a boat over deep ocean

water) pawns and items will be teleported to the nearest available cell. They used to just get deleted as they 'sunk' to the bottom of the ocean but people didn't seem to like that.

Getting Started

To start off, it's important to note that this mod is a **framework**. If you download only this mod and wonder why there is no content in your game then I hope you feel a little silly for expecting anything :)

Aside from that, when paired with vehicle mods that extend off this framework you'll notice there are quite a few mechanics aside from just adding more realistic vehicles to your game. This google doc is to give you a good idea of how to use basic vehicles from this framework, as well as what mechanics are implemented.

Features

Constructing

Vehicles are buildable through the architect menu, but which tab that will be in is going to depend on the vehicle mods you've added. By default, vehicles will be added to the Structure tab, however I presume most will create their own tab for vehicles from their mod.

Building a vehicle works just the same as any building. Select the vehicle you want to build, place it somewhere valid on the map, and then let your pawns bring materials to construct it.

Note: enabling DevMode and godMode will make vehicles build instantly just like buildings do.

Operating

Most vehicles will require at least 1 pawn to operate it. Be sure to check how many pawns will be required in order to operate a role. In order to move a vehicle, all roles responsible for movement must be fulfilled. This means the minimum count must be reached. Only pawns capable of manipulation are counted.

Vehicle components (parts) are capable of being damaged. If a part related to movement is damaged (eg. a wheel) then the overall MoveSpeed will decrease. If a critical part is damaged or enough parts are destroyed to bring the vehicle down to a crawl, the vehicle will no longer be operational until you repair all parts.

Repairing

Repairing a vehicle works the same as repairing a building. You can either wait for a pawn assigned to construction to repair the vehicle on their own accord, or you can order them through the float menu when right clicking the vehicle with the colonist selected.

Note: Repairing a vehicle will repair each vehicle part individually. This will automatically choose the highest priority part that needs to be repaired next. (Parts responsible for movement, turrets, etc.) You **can not** order a colonist to repair a specific part. The process is automatic only.

World Map Travel

Because of how involved I had to be with caravans, the dialog for forming a caravan with vehicles had to be split away from vanilla. If you select your settlement you will see 2 buttons for forming a caravan. One is "Form Vehicle Caravan" while the other is "Form caravan." If you want to form a normal vanilla caravan, choose the latter, otherwise select Form Vehicle Caravan.

When in the dialog, you'll see an extra tab that lists out all vehicles. In order to add a vehicle to the caravan, you'll need to first click the checkmark. Once doing this, the Assign Seats window will pop up. This is where you will assign seats for any potential colonists you'll be bringing with you. Any colonists that are already inside the vehicle when opening the Form Vehicle Caravan dialog will already be added in the Assign Seats window. You **cannot** remove them from their seats from this window. If you want to kick someone out of the vehicle and assign someone else for the caravan, you'll need to exit the window, go to your vehicle, and remove them there.

Once you're finished assigning colonists to your vehicle, the vehicle will be added to the caravan. All colonists assigned to this vehicle will also be added to the caravan and cannot be removed unless you remove them from the vehicle's assigned seating or remove the vehicle from the caravan entirely.

From there, the rest is the same as vanilla caravan formation.

Pathing

Unless disabled in the mod settings, all vehicles will utilize their full size for pathing. This means a 3x5 vehicle is not going to be able to path through a door. This also means you need to be more conscious about leaving space for larger size vehicles you may potentially acquire in the future.

Vehicles are also capable of defining their own path costs for just about every feature that would otherwise affect pathing. These include terrain, things, and snow for local maps as well as biomes, roads, mountainous terrain, and rivers for the world map. I highly recommend you read

any information listed in other vehicle mods' steam workshop pages for a better idea of what their pathing settings are.

Do not assume that vehicles can path to somewhere just because it would be reachable for vanilla pawns.

Note: This framework **does not** add garage doors, retractable walls, or any other building that would allow you to move a vehicle in and out of cover. You'll need to look for another mod to do this. This is a design choice as I really do want to remain within scope of the project, and currently the framework is developed strictly as a modder tool rather than a content mod.

Turrets

Vehicle turrets are similar to vanilla turrets in functionality, but entirely different to what extent you can modify them.

The most obvious difference is that vehicle turrets are not a "Thing" (RimWorld object). Essentially this means it does not occupy any space or get registered on the map. Vehicles can have *any amount* of turrets attached to them with hardpoints for where they should render / shoot projectiles from. There are quite a lot of mechanics for turrets so to keep it short I'll describe the more impactful ones.

- They can be given limited traverse angles. This will prevent the turret from rotating, aiming, and firing at an angle that is beyond its sector of fire.
- They can be attached to other turrets. One example of this would be a mounted machine gun on a tank. Attached turrets will rotate with their parent turret and adjust their sector of fire accordingly.
- They have what I call "Fire Modes." It essentially defines the rate of fire of the turret. It can range anywhere from semi-auto to full-auto. This is mostly for modders, as this allows them to define and balance the turrets to what they feel is right. They **cannot** be modified in-game through the mod settings, however modders can give a turret more than 1 fire mode, and clicking the relevant button on the in-game turret gizmo will toggle through the different fire modes.
- They have custom projectile flags that can be applied at runtime. This won't really matter to you the player, however it's something to look out for as vehicles can define their own behavior for projectiles launched from their turrets. eg. I have a tank for testing that fires mortar shells at ~150c/s that will fly over almost everything, but still hit buildings and other vehicles.

Paint Jobs

This is one of my favorites of the listed features. I've created 3 shaders that provide some very awesome customization. The first is an RGB shader that behaves the exact same as vanilla with an extra color channel allowing for 3 colors for masking instead of 2. The second is the RGBPattern shader that takes this way further and applies a pattern to the mask instead of flat

colors. You heard that right, you can paint your vehicles with camo, stripes, spots, hearts, or whatever else you have available in the ColorPicker window. The last of the shaders is the RGBSkin shader which is the exact same concept as RGBPattern except it applies the texture directly without color masking. If you have some elaborate design you would like to overlay onto the vehicle without having to worry about converting it to Red / Green / Blue channels, you can do so with this last one.

To change a vehicle's color and pattern you first need to make sure the vehicle supports this. I cannot do anything with vehicles that have not assigned my shader to their vehicle. You can check this in-game by selecting your vehicle and looking for a paintbrush icon in the inspect pane (bottom left window). If it has a paintbrush then you're good to go! Just click it and it will pop up the window with all applicable patterns and skins along with a color picker to the right. You also have access to a ton of color palettes for saving your favorite color presets.

Mod Settings

Main

The main page for any settings that apply in a general way. If the setting is in this section, that means it will apply across your entire game (and to every applicable vehicle).

Vehicles

This is where you can tweak each vehicle's settings to suit your own tastes. Each field has 3 possible states:

Available: You can edit this field as much as you want. If the field has been edited, it will be colored blue indicating that it will write to your config file, disallowing any future updates from changing this value. If it is colored white, it will use whatever default value comes from the xml of the vehicle.

Locked: Blocked from editing due to not being applicable to this vehicle (eg. fishing for land vehicles) or the modder has decided they do not want players changing this field.

Disabled: feature / mechanic does not currently work and thus cannot be modified.